Final_Paper

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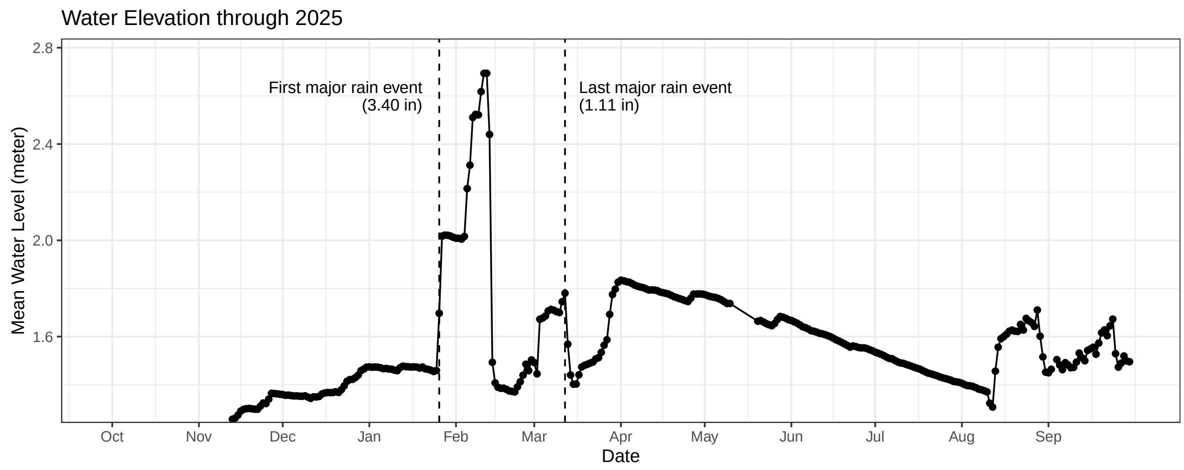
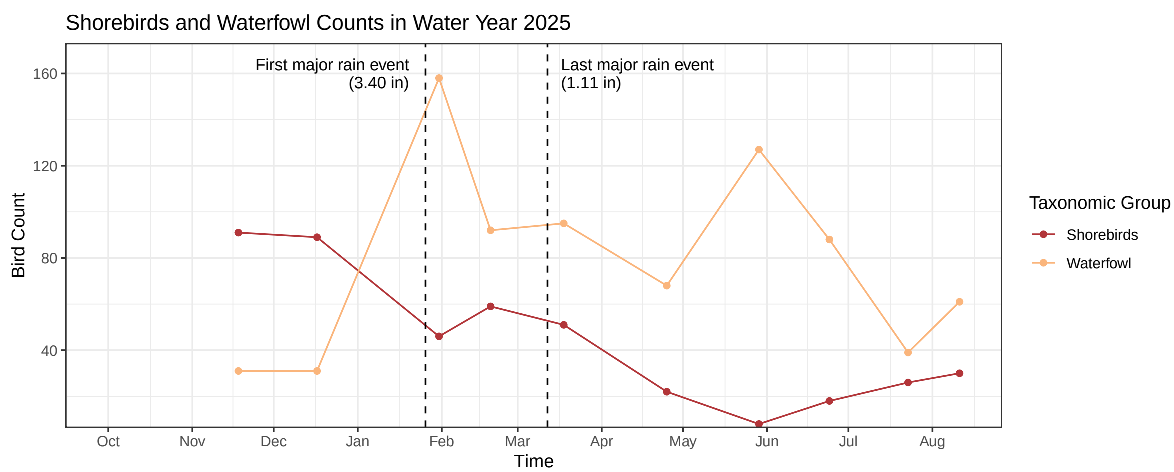
First, we explored water elevation and taxonomic abundance as separated variables over time. The first plot shows water elevation through time in the 2025 water year, along with major rain events that significantly spiked water elevation. A limitation for this data is that this water elevation is accurate to local Venoco bridge water elevation, not necessarily the whole wetland.



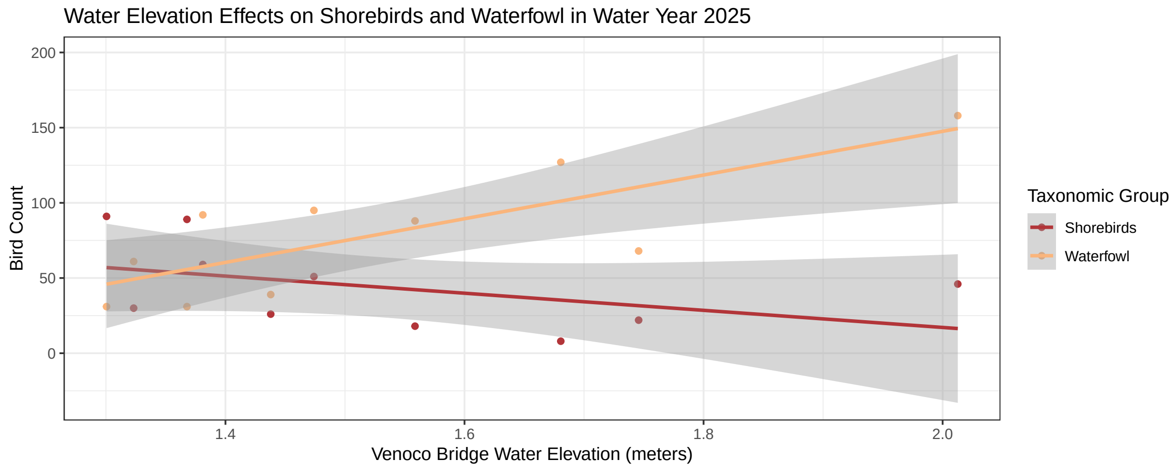
Here you can observe that the highest elevation reached around 2.6 meters with highest elevations concentrating in February 2025. After April, the water levels exhibit a steady decline until August. We then examined how bird counts looked over time within the two taxonomic groups Waterfowl and Shorebirds.



Around the February, Waterfowl showed the highest abundance, adapting well to the high water elevation while its counterpart, the Shorebirds, showed a heavy decline in abundance from high water elevation. Shorebirds continued to decline in numbers, showing a record low right before June before rising back up through to August as water elevation continued to decline.

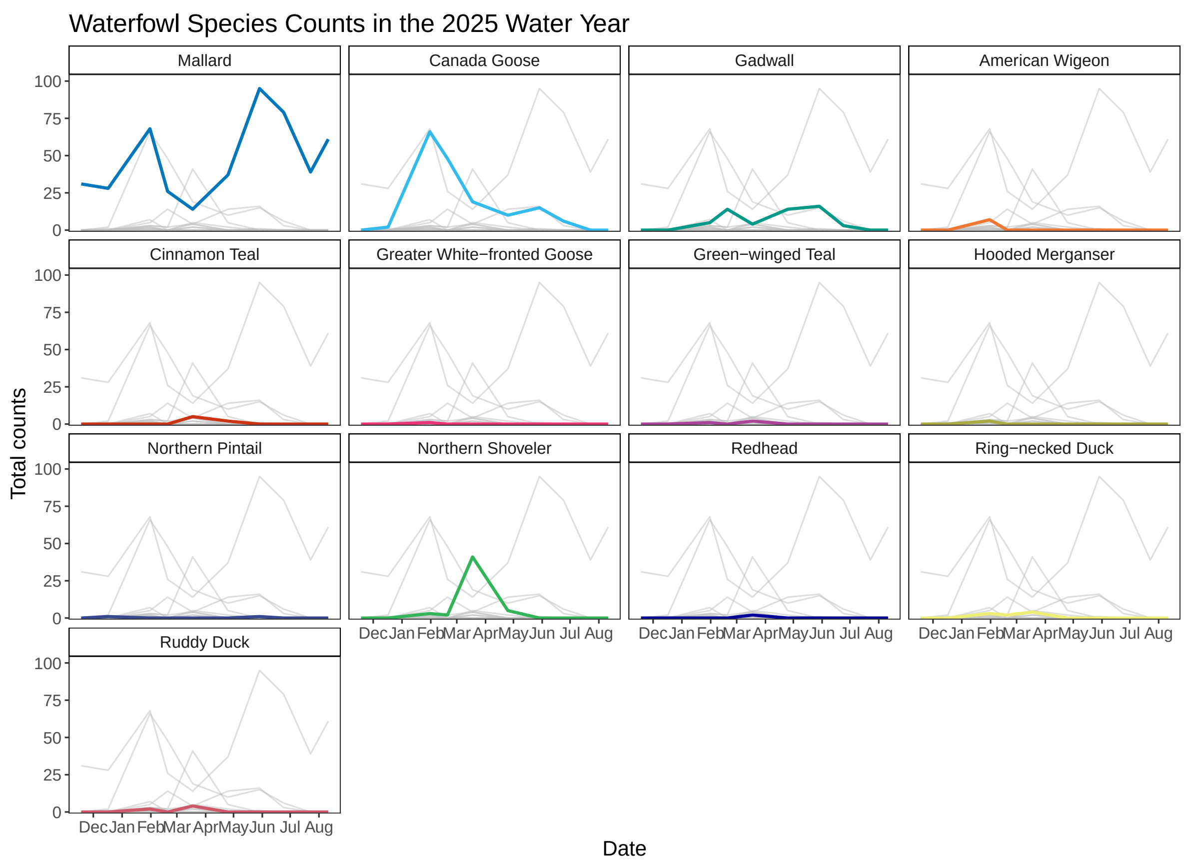
After reaching their record highest abundance in February, Waterfowl began to decline in March, but suddenly shot back up in abundance in June when Shorebird abundance was almost zero. The plot highlights sharp deviation between the two taxonomic groups, revealing how water elevation affects them differently.

After plotting each variable separately, we continued our exploration by creating a scatter plot combining both variables.



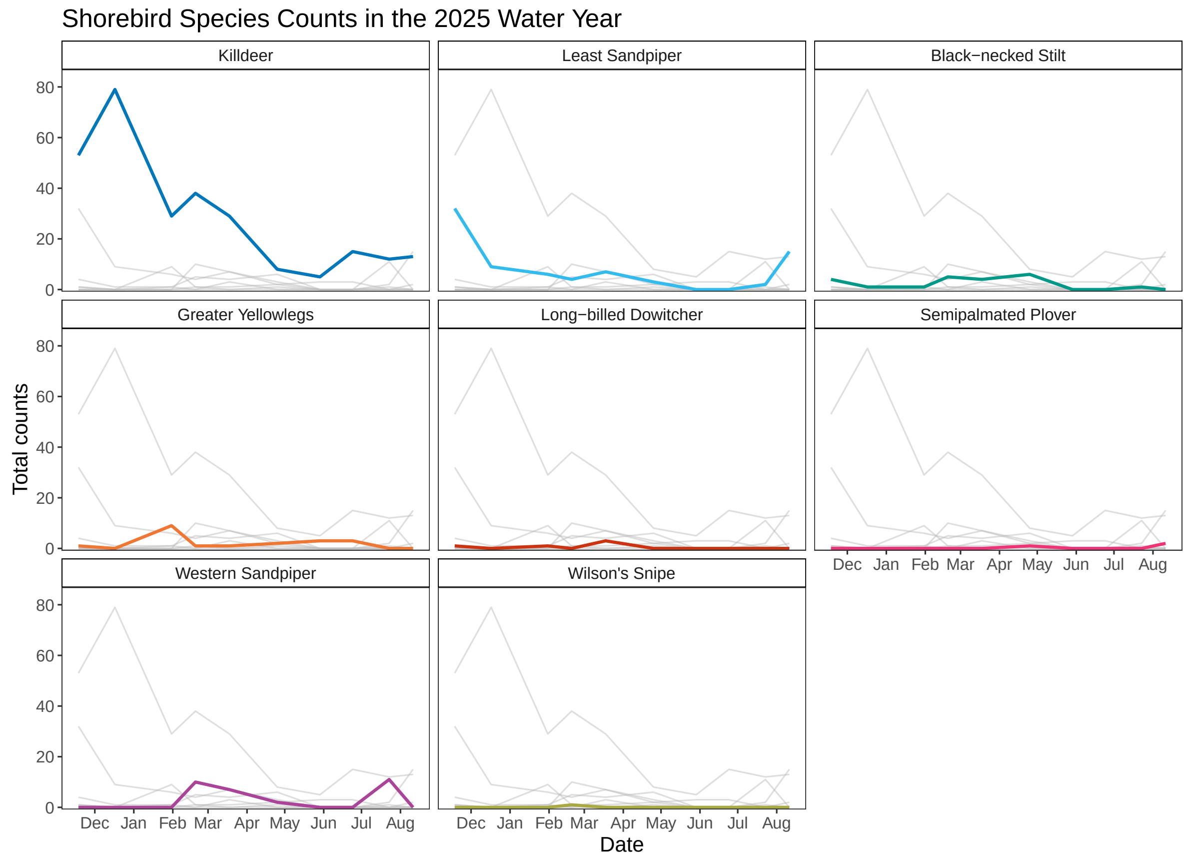
This plot revealed correlation between water elevation and these taxonomic groups in NCOS. We applied the Pearson's correlation test to find exact values of this correlation. Waterfowl show a positive correlation between bird abundance and water elevation with a correlation value of 0.785. In contrast, Shorebirds have a negative correlation value of -0.445. Therefore, this scatter plot shows how as water elevation increases, Shorebird abundance drops, and Waterfowl abundance climbs.

These plots only revealed patterns for each taxonomic group as a whole, therefore we decided to plot species abundance over time to see which species were exhibiting this correlation the most. We chose highlight plots to visualize abundance through time for each species.



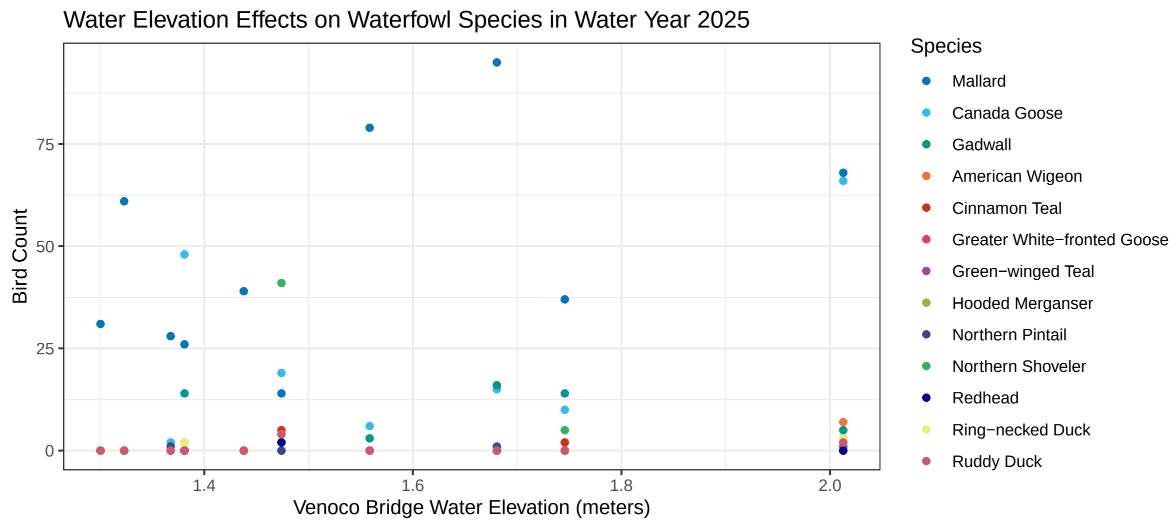
Upon further inspection of waterfowl individual species, the general taxonomic trends are attributed to the two most abundant species of waterfowl.The Mallard and Canadian Goose makeup the general taxonomic trends, meaning the positive correlation of 0.785 wouldn't apply to the 11 remaining waterfowl species.

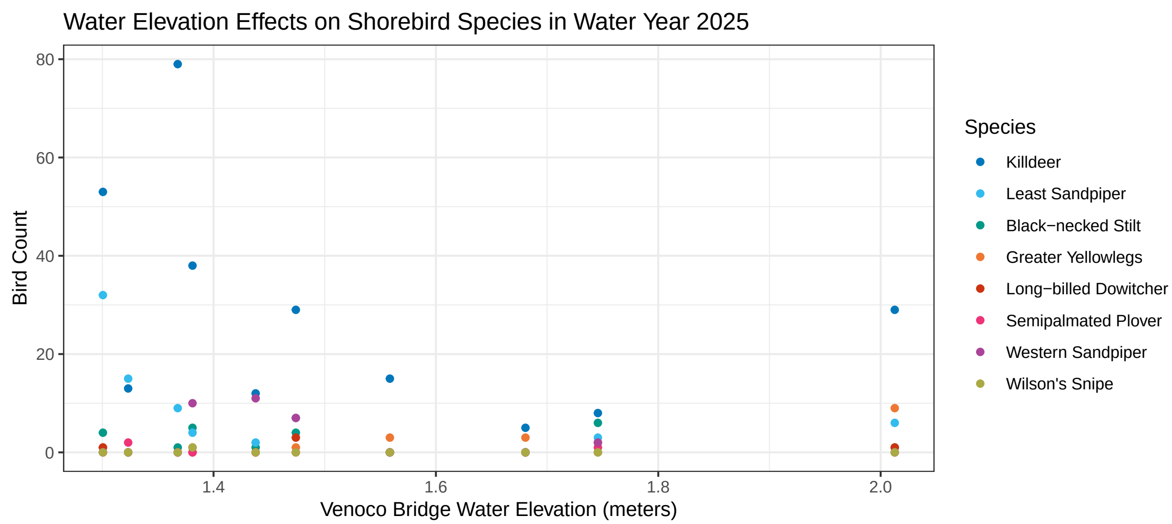
We applied the same test to Shorebirds.



The same phenomenon occurs with the Shorebirds as the two most abundant species, the Killdeer and Least Sandpiper, follow the negative correlation trend the closes - especially the Killdeer. The remaining 6 Shorebird species did not closely follow this correlation at all.

Finally, we combined individual species abundance with water elevation to see how the most abundant species were affected by different water levels.





These final figures display how only the most abundant species display the general correlation trends. Therefore examining bird abundance by taxonomic groups yields inaccurate patterns, since the data is highly influenced by only a select few species, rather than the group as a whole. Water elevation effects on bird species should be evaluated by species to address nuance between each species within the taxonomic group.

In conclusion, our results highlight how the Mallard and Canadian Goose (Waterfowl) are more abundant in water elevations around 2 meters while the Killdeer and Least Sandpiper (Shorebirds) are more abundant in water elevations below 1.4 meters.

Discussion

References